

Rain Hu

Senior Backend Engineer

📍 Taiwan

in [LinkedIn](#)

☎ 0931-639-433

✉ intervalrain@gmail.com

🐙 [GitHub](#)

PROFESSIONAL SUMMARY

Senior backend engineer with 7+ years in the semiconductor industry. Experienced in designing scalable backend systems using .NET Core/ASP.NET Core, Clean Architecture, andDDD. Proven ability to build containerized microservices, implement CI/CD pipelines, and apply TDD/12-factor app principles. Delivered REST APIs, AI-integrated services, and cross-site enterprise platforms for semiconductor manufacturing.

CORE COMPETENCIES

Languages: C#, TypeScript, C++, Golang, Python,
Architecture: Clean Arch, DDD, CQRS, Microservices
Practices: TDD, CI/CD, Scrum, Agile
Databases: PostgreSQL, Oracle, SQLite

Backend: .NET Core, Node.js, FastAPI
Containers: Docker, Docker Compose
DevOps: Azure DevOps, Git, Grafana, GitHub, GitLab
AI: LangChain, RAG, Claude Code, Skills, ML

PROFESSIONAL EXPERIENCE

Assistant Manager / Senior Software Engineer
Advantech

02/2025 – Present
Taoyuan, Taiwan

- Architected digital-twin services (Shadow Agent, Transceiver, FileTransfer) for edge-cloud synchronization, enabling device state reconciliation, telemetry processing, and chunked AI model delivery to secured volumes.
- Built Container Management application on top of digital-twin platform, supporting versioned rollout, staged release, and remote lifecycle management across edge devices.
- Developed ServiceFramework SDK with NATS-based RPC, retry policies, dead-letter queue, and transactional outbox pattern for reliable distributed microservice communication.
- Established CI/CD pipelines (Azure DevOps) with automated testing (unit/integration); integrated Grafana for monitoring and observability.
- Applied DDD, Clean Architecture, and CQRS (Mediator) to decouple domain logic; enforced TDD and code review practices.
- Open-sourced SubNode SDK supporting Modbus TCP/RTU, MQTT, and OPC UA for third-party devices to connect WEDA Core IoT platform.

Senior Software Engineer
UMC (United Microelectronics Corp.)

08/2022 – 10/2024
Tainan, Taiwan

- Led Clean Architecture adoption for UEDA 5.0, decoupling domain logic from infrastructure to support cross-fab schema heterogeneity across global sites (Taiwan, Japan, Singapore, China).
- Developed new analytics modules (Lot History Alignment using SCS algorithm, BTV Analysis, DSM Compare) and refactored legacy modules (Wafermap, Bin-to-Defect, Defect Kill Ratio, etc).
- Built DSM Bot using LangChain.NET and Azure OpenAI with RAG; enforced LDAP-based access control over secured documents(DSM) before retrieval and response generation.
- Optimized database queries across Oracle, PostgreSQL, and Hadoop, improving query performance by 40% for semiconductor analytics.

Device R&D Engineer
UMC (United Microelectronics Corp.)

08/2018 – 08/2022
Hsinchu, Taiwan

- Developed eHV device processes across 22nm/28nm HKMG and 40nm Poly-Gate nodes for display IC and OLED applications.
- Designed algorithmic mask layout optimization (NWR) improving yield by 5%; adopted as BKM across both eHV and logic platforms.
- Built EDA tools (U2C DRC translator, layout pipelines) in Python/C++, reducing engineering workload by 30–50%.
- Performed WAT analysis, test-key layout design, and device characterization tuning across multiple technology nodes.

EDUCATION

MS, Material Science & Engineering
National Tsing Hua University
BS, Material Science & Engineering
National Tsing Hua University

2014 – 2018
GPA: 3.97
2010 – 2014
GPA: 3.61